

BLOCKCHAIN TECHNOLOGIES 2 — OPTION A

DEFI SUPER-APP

AMM · LENDING · ERC-4626 VAULT
DAO GOVERNANCE · CHAINLINK · THE GRAPH

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[GITHUB.COM/ADILZHANKAD/BCH2_FINAL](https://github.com/adilzhankad/bch2_final)

11

SMART CONTRACTS

164

TESTS PASSING

94.85%

LINE COVERAGE

L2

OPTIMISM SEPOLIA

WHAT WE BUILT

6 CORE CONTRACTS

- **AMM** (X·Y=K, 0.3% FEE)
- **LENDINGPOOL** (LTV/HF)
- **YIELDVAULT** (ERC-4626)
- **POOLFACTORY** (CREATE2)
- **PRICEORACLE** (CHAINLINK)
- **TREASURY**

3 GOVERNANCE

- **GOVTOKEN** (ERC20VOTES)
- **DEFIGOVERNOR** (1D/7D/4%)
- **TIMELOCKCONTROLLER** (2D)
- **PROTOCOLNFT** (ERC-721)

FRONTEND

- **NEXT.JS 14 + TYPESCRIPT**
- **WAGMI V2 + RAINBOWKIT**
- **4 PAGES** (SWAP/VAULT/LENDING/GOVERNANCE)
- **NETWORKGUARD**
- **TXBUTTON W/ ERRORS**

SUBGRAPH

- **15 ENTITIES**
- **7 MAPPINGS**
- **5+ GRAPHQL QUERIES**
- **REAL-TIME INDEXING**
- **OPTIMISM SEPOLIA**
- **EVENT-DRIVEN**

TESTING

- **164 TESTS TOTAL**
- **94.85% LINE COVERAGE**
- **130+ UNIT TESTS**
- **13 FUZZ TESTS**
- **13 INVARIANT TESTS**
- **4 FORK TESTS**

OUR APPLICATION

HomeSwapVaultLendingGovernance

0.039 ETH

DeFi Super-App

AMM · Lending · ERC-4626 Vault · DAO Governance

YOUR WALLET

DGT Balance

9999898.00 DGT

Voting Power

9999898.00 DGT

AMM POOL

Reserve 0

—

Reserve 1

—

YIELD VAULT

TVL

102.00

Swap

Trade tokens

Vault

Earn yield

Lending

Borrow & lend

Governance

Vote on proposals

Swap

Pool Reserves

Token 0

—

Token 1

—

Token0 → Token1

Flip

You Pay Balance: —

0.0

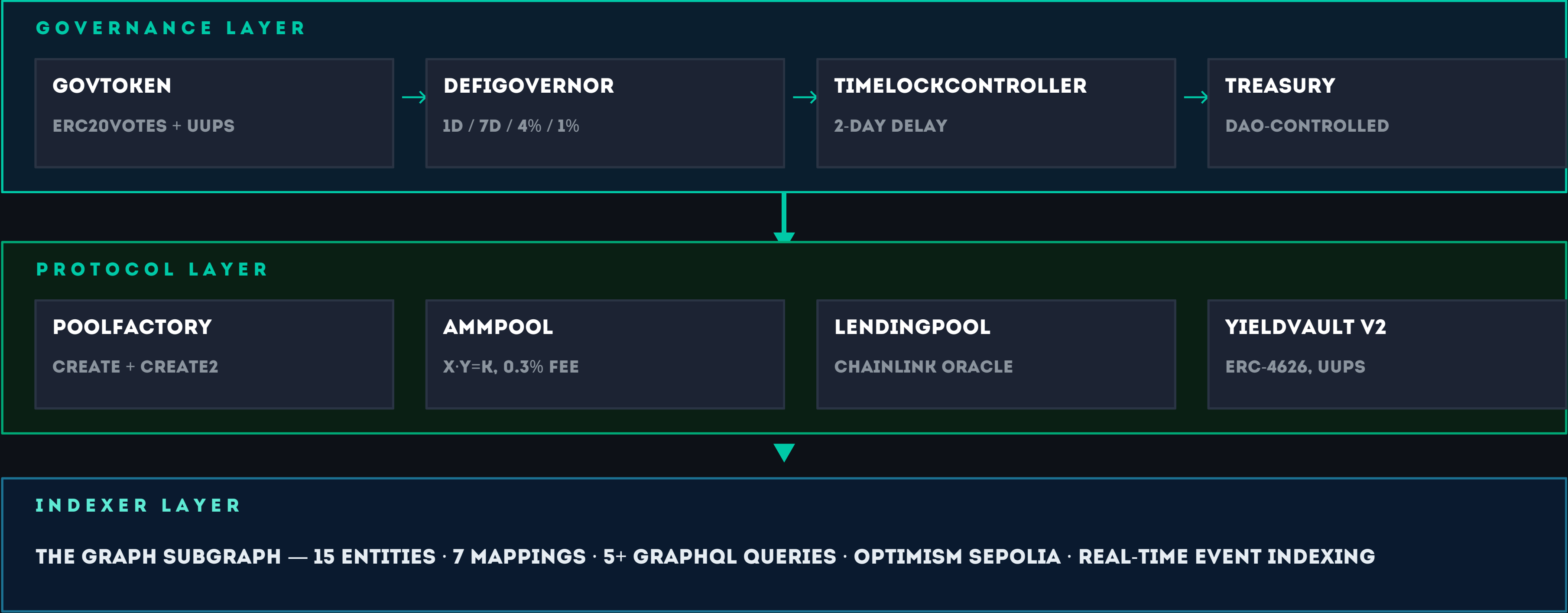
You Receive (est.)

—

Swap

SYSTEM ARCHITECTURE

THREE LAYERS, ZERO ADMIN BACKDOOR — ALL PRIVILEGED ACTIONS FLOW THROUGH DAO WITH 2-DAY TIMELOCK



TECH STACK

SMART CONTRACTS

SOLIDITY 0.8.24

FOUNDRY (FORGE)

OPENZEPPELIN V5

SLITHER CI

FORGE FMT + SOLHINT

FRONTEND

NEXT.JS 14 (APP ROUTER)

TYPESCRIPT 5

RAINBOWKIT + WAGMI V2

VIEM

TANSTACK QUERY

INFRASTRUCTURE

OPTIMISM SEPOLIA (L2)

ETHERSCAN VERIFICATION

THE GRAPH SUBGRAPH

GITHUB ACTIONS CI

FOUNDRY-TOOLCHAIN

REQUIREMENTS COVERAGE

EVERY MANDATORY DOMAIN FROM THE SPECIFICATION — FULLY IMPLEMENTED, NO STUBS

✓	ADVANCED SOLIDITY	UUPS UPGRADE V1→V2 (GOVTOKEN + YIELDVAULT), POOLFACTORY CREATE+CREATE2, INLINE YUL SORT BENCHMARK
✓	TOKENS (ERC STANDARDS)	ERC20VOTES + PERMIT, ERC-721 PROTOCOLNFT, ERC-4626 VAULT WITH ROUNDING INVARIANTS
✓	DEFI PRIMITIVES	CONSTANT-PRODUCT AMM FROM SCRATCH, LENDINGPOOL WITH LTV / HF / LIQUIDATION / LINEAR RATE
✓	ORACLE INTEGRATION	CHAINLINK AGGREGATORV3 + STALENESS GUARD + NEGATIVE-ANSWER + INCOMPLETE-ROUND CHECKS
✓	GRAPH INDEXING	SUBGRAPH: 15 ENTITIES, 7 MAPPINGS, 5+ GRAPHQL QUERIES, REAL-TIME EVENT INDEXING
✓	DAO GOVERNANCE	FULL OZ STACK: 1D DELAY, 7D PERIOD, 4% QUORUM, 1% THRESHOLD, 2-DAY TIMELOCK
✓	LAYER 2 DEPLOYMENT	DEPLOYED + VERIFIED ON OP SEPOLIA, L1 VS L2 GAS TABLE — 96-97% CHEAPER

AMM

CONSTANT-PRODUCT · 0.3% FEE · LP TOKENS

CONSTANT PRODUCT

X·Y=K FORMULA, INVARIANT VERIFIED WITH 1,024 FUZZ CALLS

SWAP FUNCTIONS

SWAPEXACTIN / SWAPEXACTOUT WITH SLIPPAGE PROTECTION (MINAMOUNTOUT)

FEE STRUCTURE

0.3% FEE VIA 997/1000 MULTIPLIER, LP TOKEN MINTING $\text{SQRT}(A0 \times A1)$

YUL OPTIMIZATION

INLINE ASSEMBLY SQRT: 25,867 → 10,864 GAS (58% SAVING)

```
// INLINE YUL SQRT — BABYLONIAN
FUNCTION _SQRT(UINT256 X)
  INTERNAL PURE
  RETURNS (UINT256 Y) {
  ASSEMBLY {
    // ~10K GAS VS ~25K SOLIDITY
  }
}
```

58%

GAS SAVED
(YUL SQRT)

$K \geq K$

INVARIANT
1,024 FUZZ CALLS

LENDINGPOOL

OVER-COLLATERALIZED BORROWING WITH ON-CHAIN PRICE ORACLE

75%

LTV RATIO

80%

LIQ. THRESHOLD

5%

LIQ. BONUS

35+

UNIT TESTS

● **LINEAR INTEREST RATE, ACCRUED PER SECOND VIA BLOCK.TIMESTAMP DELTA**

● **SEPARATE ACCOUNTING: TOTALCOLLATERAL VS TOTALLIQUIDITY (TWO MAPPINGS)**

● **PARTIAL REPAY RETURNS COLLATERAL PROPORTIONALLY (FIXED BUG)**

● **ZERO ADMIN BACKDOOR — ALL PRIVILEGED ACTIONS THROUGH DAO + 2D TIMELOCK**

● **HEALTH FACTOR COLOR-CODED UI: RED <1.1 / YELLOW <1.5 / GREEN ≥ 1.5**

● **35 UNIT + 2 FUZZ + INVARIANT TESTS — NO REENTRANCY ISSUES FOUND**

YIELDVAULT (ERC-4626)

UUPS UPGRADEABLE

V1 → V2 UPGRADE PATH (EIP-7201 STORAGE)

FULL ERC-4626

DEPOSIT / MINT / WITHDRAW / REDEEM

YIELD INJECTION

INJECTYIELD RAISES SHARE PRICE FOR ALL HOLDERS

PAUSABLE

PAUSER_ROLE ONLY — VIA DAO GOVERNANCE

V2 FEATURE

PERFORMANCE FEE ON YIELD HARVESTED

DAO GOVERNANCE



PROPOSAL THRESHOLD

1% OF TOTAL GOVTOKEN SUPPLY

TIMESTAMP CLOCK

FIXED FOR L2 (BLOCK-NUMBER BROKEN ON OP)

EXECUTION FLOW

PROPOSE → VOTE → QUEUE → EXECUTE

SECURITY

FLASH-LOAN PROTECTED BY SNAPSHOT + 1D DELAY

END-TO-END

FULL LIFECYCLE TESTED IN FORK TESTS

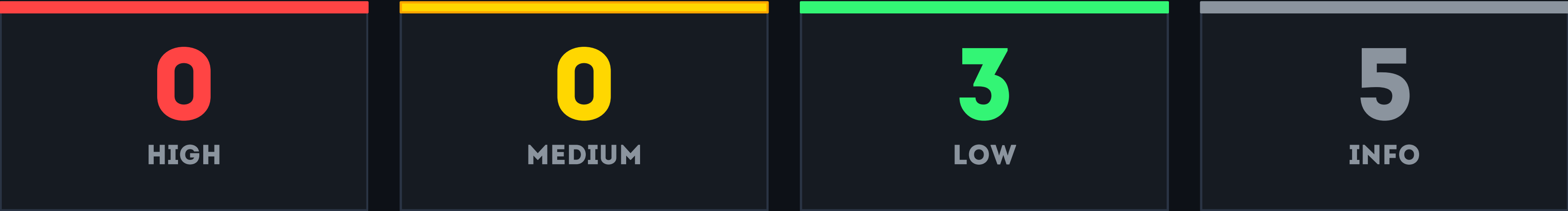
164 TESTS · 94.85%

LINE COVERAGE · EVERY CONTRACT ABOVE 90%



✓	TREASURY ACCOUNTING INVARIANT	REQUIRED BY SPEC §3.3 — 1024 CALLS, 0 VIOLATIONS
✓	ERC-4626 ROUNDING INVARIANTS	CONVERTTOSHARES / CONVERTTOASSETS ROUND-TRIP VERIFIED
✓	AMM CONSTANT-PRODUCT	K NEVER DECREASES ON ANY SWAP — PROVEN BY FUZZ
✓	FORK TESTS ON REAL CHAIN	CHAINLINK ETH/USD FEED ON MAINNET BLOCK 19,000,000

INTERNAL SECURITY AUDIT



DEFENCES ANALYSED

SLITHER IN CI	--FAIL-HIGH --FAIL-MEDIUM — BUILD RED ON ANY FINDING
MANUAL REVIEW	LINE-BY-LINE REVIEW BY BOTH TEAM MEMBERS
PROPERTY TESTS	13 FUZZ + 13 INVARIANT TESTS WITH HANDLER-BASED STATEFUL RUNS
CENTRALIZATION	DEPLOYER = 0 ROLES AFTER HANDOFF — VERIFIED IN CI (21 INVARIANTS)
ORACLE GUARDS	STALE / NEGATIVE / INCOMPLETE-ROUND CHECKS — 3 GUARD CONDITIONS

GAS OPTIMIZATION

ALL NUMBERS FROM FORGE --GAS-REPORT — FULLY REPRODUCIBLE

OPTIMIZATION	BEFORE	AFTER	SAVED
_SORT (SOLIDITY → YUL)	25,867	10,864	58%
UUPS VS TRANSPARENT PROXY	~2,100 / CALL	0 / CALL	~2,100
LENDINGPOOL ACCOUNTING SPLIT	~5,400 / BORROW	~3,200 / BORROW	~40%

96–97%

CHEAPER

L1 VS L2 COST COMPARISON

MEASURING 6 OPERATIONS: SWAP, ADDLIQUIDITY, BORROW, LIQUIDATE, VAULT DEPOSIT, CASTVOTE
OPTIMISM SEPOLIA IS 30-100× CHEAPER PER TRANSACTION THAN ETHEREUM MAINNET

FRONTEND & SUBGRAPH

SWAP

TOKEN EXCHANGE WITH SLIPPAGE
CONTROL

VAULT

DEPOSIT / WITHDRAW ERC-4626
VAULT

LENDING

BORROW, REPAY, HEALTH FACTOR
UI

GOVERNANCE

VOTE, DELEGATE, PROPOSE
CHANGES

FRONTEND (NEXT.JS + WAGMI V2)

- RAINBOWKIT — METAMASK + WALLETCONNECT
- NETWORKGUARD — AUTO-PROMPT NETWORK SWITCH
- TXBUTTON WITH HUMANIZED ERRORS (REJECTED / REVERT)
- 3+ WRITE TX: SWAP, DEPOSIT, VOTE, DELEGATE, BORROW

SUBGRAPH (THE GRAPH)

- 15 ENTITIES ACROSS ALL CONTRACTS
- 7 MAPPINGS (AMM, LENDING, VAULT, GOVERNOR, NFT)
- 5+ GRAPHQL QUERIES IN SRC/LIB/SUBGRAPH.TS
- ACTIVE PROPOSALS READ FROM SUBGRAPH, NOT CONTRACT

OPTIMISM SEPOLIA DEPLOYMENT

ALL CONTRACTS VERIFIED ON ETHERSCAN · DEPLOYER HOLDS ZERO ROLES AFTER HANDOFF

CONTRACT	ADDRESS	STATUS
POOLFACTORY	0X854B...E690	✓ VERIFIED
LENDINGPOOL	0X8A79...7141	✓ VERIFIED
YIELDVAULT	0XC391...0428	✓ VERIFIED
DEFIGOVERNOR	0XCC26...3ECA	✓ VERIFIED
PROTOCOLNFT	0X6E84...BC62	✓ VERIFIED

DEPLOYMENT PROCESS

- 1

DEPLOY.S.SOL
IDEMPOTENT, REPRODUCIBLE SINGLE SCRIPT
- 2

_HANDOFFTOTIMELOCK
ALL ADMIN ROLES → TIMELOCK, DEPLOYER REVOKED
- 3

VERIFY.S.SOL
VERIFICATION SCRIPT POST-DEPLOY
- 4

21 CI INVARIANTS
CHECKED ON EVERY COMMIT VIA DEPLOYVERIFY.T.SOL

THANK YOU.

READY FOR QUESTIONS.

LINK

**[GITHUB.COM/ADILZHANKAD/
BCH2_FINAL](https://github.com/adilzhankad/bch2_final)**

ETHERSCAN

**SEPOLIA-
OPTIMISM.ETHERSCAN.IO**

CI

**164 TESTS PASSING ON
EVERY COMMIT**

DOCS

DOCS/FULL-REPORT.DOCX